

CAT, the Node with Node ID 0x6666_6666_6666_6666 will again have access to Nodes that list subject 0xFFFF_FFFD_ABCD_0002 (CAT identifier value 0xABCD, version 0x0002), with no further updates to ACL entries of existing Nodes.

- Any controlled Node which previously held an ACL Entry with prior version of the updated CAT (subject 0xFFFF_FFFD_ABCD_0001) but was not reachable by an Administrator at the time of update, will continue to hold the previous Access Control Entry with a subject allowing CAT with identifier of 0xABCD and version 0x0001 or higher. Thus, these Nodes will grant privileges to any Node from the original CAT group (including Node ID 0x3333_3333_3333_3333). When an Administrator eventually establishes connection to this Node with older ACL entry, the Administrator SHOULD update it with the latest value, so that Node ID 0x3333_3333_3333_3333 no longer has privileges.

Note that in the above example, the CAT identifier value remained the same (0xABCD) in NOCs and ACL entries throughout these steps. Only the version portion was updated to effect changes to the meaning of the CAT.

As can be seen in the example above, there are multiple steps involving updates to NOCs and ACL entries to affect CAT-based grouping and aliasing policies. It is therefore possible that some Nodes may not receive these changes immediately, due to network reachability issues, such as being powered down for an extended period, and thus have ACL entries or NOCs that grant temporarily obsolete privileges. This is true as well with direct Node ID subjects, in general.

Administrators SHOULD aim for best-effort eventual consistency while executing the steps outlined above.

```
Access Control Cluster: {
  ACL: [
    ...
    3: {
      FabricIndex: 1,
      Privilege: Operate,
      AuthMode: CASE,
      Subjects: [ 0x1111_1111_1111_1111, 0xFFFF_FFFD_ABCD_0002 ],
      Targets: [ { DeviceType: 0x0000_010D } ]
    }
  ]
}
```

6.6.4. Access Control Cluster update side-effects

Updates to the Access Control Cluster SHALL take immediate effect in the Access Control system. For example, given an Interaction Model action message containing the following actions, the Access Control Privilege Granting algorithm would grant a privilege of **None** for the second action, since the first action would take effect immediately beforehand.

- Pre-conditions: